

Companion XIV: Primordial Black Holes as CPT-Sector Objects in the Relaxation-Driven Cyclic Cosmology (RDCC)

Michael Lehmann¹

¹Independent Researcher, Relaxation-Driven Cyclic Cosmology (RDCC) Framework,
mi.lehmann@gmx.de

May 15, 2026

Abstract

Primordial black holes (PBHs) provide a unique probe of early-Universe physics, sensitive to the thermal history, the growth of perturbations, and the collapse thresholds during radiation domination. In the Relaxation-Driven Cyclic Cosmology (RDCC), the bipartite CPT-symmetric global state and the universal relaxation mechanism fundamentally alter the conditions for PBH formation. In this Companion we develop a new interpretation of PBHs within RDCC: PBHs arise not in the visible sector, but in the CPT-conjugate sector, where the lower temperature, modified relaxation history, and altered perturbation dynamics naturally enhance PBH formation. The visible sector, by contrast, exhibits a strong suppression of PBHs due to relaxation-induced damping and an elevated collapse threshold. We derive the CPT-sector PBH formation mechanism, compute the resulting mass function, and identify the observational signatures that arise from this sectoral asymmetry. This Companion establishes PBHs as a natural and robust component of the CPT sector and provides a new observational window into the bipartite structure of RDCC.

1 Introduction

Primordial black holes (PBHs) provide a unique observational window into the earliest stages of cosmic evolution. Their formation depends sensitively on the thermal history, the growth of primordial perturbations, and the collapse thresholds during radiation domination. In standard cosmology, PBHs require either enhanced primordial fluctuations, non-Gaussian features, or exotic early-Universe dynamics to form in appreciable numbers.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is fundamentally different. The bipartite CPT-symmetric global state and the universal relaxation mechanism modify the early-Universe dynamics in ways that directly impact PBH formation. The relaxation rate alters the growth of super-horizon perturbations, increases the effective collapse threshold, and introduces a sectoral asymmetry due to the temperature difference between the two sectors.

In this Companion we develop a new interpretation of PBHs within RDCC: PBHs arise not in the visible sector, but in the CPT-conjugate sector. The colder temperature, modified relaxation history, and altered perturbation dynamics in the CPT sector naturally enhance PBH formation, while the visible sector experiences a strong suppression. This sectoral asymmetry is a robust and testable prediction of the RDCC framework.

The structure of this Companion is as follows. Section 2 introduces PBHs as CPT-sector objects and explains why PBH formation is naturally suppressed in the visible sector. Section 3 derives the CPT-sector PBH formation mechanism. Section 4 develops the resulting PBH

mass function. Section 5 analyzes the thermodynamic and dynamical properties of CPT-sector PBHs. Section 6 presents the observational signatures, and Section 7 provides forecasts for future surveys.

2 PBHs as CPT-Sector Objects

The RDCC framework is built upon a bipartite, CPT-symmetric global state in which two thermodynamically distinct sectors evolve on the same spacetime geometry. The observable Universe corresponds to a projection onto the sector with a forward thermodynamic arrow (sector A), while the CPT-conjugate sector (sector B) evolves with decreasing entropy. The two sectors interact only through gravity and the relaxation dynamics.

A key result of Companion X is that the temperature ratio between the sectors satisfies

$$\frac{T_B}{T_A} \approx 0.5, \quad (1)$$

implying that the CPT-conjugate sector is colder throughout the early Universe. This temperature asymmetry has profound implications for PBH formation.

PBH formation is highly sensitive to both the collapse threshold and the horizon mass at the time of collapse. In the visible sector, the relaxation-induced damping of super-horizon perturbations increases the effective collapse threshold, suppressing PBH formation. In the CPT-conjugate sector, the lower temperature and reduced relaxation-induced damping lead to a lower collapse threshold and a larger horizon mass, enhancing PBH formation.

This leads to three robust conclusions:

- (a) **PBHs are strongly suppressed in the visible sector.** The relaxation-induced enhancement of the collapse threshold prevents overdensities from reaching the required amplitude for collapse.
- (b) **PBHs form naturally in the CPT-conjugate sector.** The colder temperature and reduced damping lower the collapse threshold and increase the horizon mass, making PBH formation efficient.
- (c) **PBHs are inherently CPT-sector objects in RDCC.** Their formation mechanism, abundance, and mass spectrum are determined by the thermodynamic and relaxation properties of sector B.

This interpretation resolves several conceptual tensions. It explains the absence of PBHs in the visible sector without fine-tuning, provides a natural origin for PBHs in the RDCC framework, and connects PBH physics directly to the bipartite structure of the global state.

In the next section, we derive the CPT-sector PBH formation mechanism in detail and quantify the degree of suppression in the visible sector.

3 CPT-Sector PBH Formation Mechanism

Primordial black holes (PBHs) form when large density fluctuations re-enter the horizon during radiation domination and exceed a critical collapse threshold. In standard cosmology, this threshold is of order $\delta_c \sim 0.4\text{--}0.5$, and PBH formation requires enhanced primordial fluctuations or non-Gaussian features.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the early-Universe dynamics are modified by the universal relaxation mechanism introduced in Companion X. The relaxation rate $\Gamma(a)$ alters the growth of super-horizon perturbations and increases the effective collapse threshold in the visible sector. As a result, PBH formation in sector A is strongly suppressed.

In the CPT-conjugate sector (sector B), the situation is fundamentally different. The lower temperature, reduced relaxation-induced damping, and altered perturbation dynamics naturally enhance PBH formation. The key ingredients are:

- (a) **Reduced relaxation-induced damping.** The relaxation coefficient satisfies $\alpha_B(a) < \alpha_A(a)$ due to the lower temperature of sector B. This reduces the damping of super-horizon perturbations and lowers the effective collapse threshold.
- (b) **Larger horizon mass.** The colder temperature implies a slower expansion rate and a larger horizon mass at a given scale factor:

$$M_H^{(B)} \approx 4M_H^{(A)}. \quad (2)$$

- (c) **Lower collapse threshold.** The reduced pressure in the colder sector lowers the collapse threshold:

$$\delta_c^{(B)} < \delta_c^{(A)}. \quad (3)$$

Combining these effects yields a natural and robust PBH formation mechanism in the CPT sector. PBHs arise as a direct consequence of the bipartite global state and the relaxation dynamics, without requiring enhanced primordial fluctuations or exotic early-Universe physics.

In contrast, the visible sector experiences a strong suppression of PBH formation due to the elevated collapse threshold and reduced horizon mass. This sectoral asymmetry is a distinctive and testable prediction of RDCC.

4 Modified Collapse Threshold in RDCC

Primordial black holes form when large density fluctuations re-enter the horizon during radiation domination and exceed a critical collapse threshold. In standard cosmology, this threshold is of order $\delta_{\text{std}} \sim 0.4\text{--}0.5$. In the Relaxation-Driven Cyclic Cosmology (RDCC), the early-Universe dynamics are modified by the universal relaxation mechanism introduced in Companion X, which alters the growth of super-horizon perturbations and increases the effective collapse threshold in the visible sector.

Relaxation-Induced Damping from Companion X and VII

The evolution of super-horizon perturbations in RDCC follows a modified equation of motion,

$$\ddot{\delta} + (2H + \Gamma) \dot{\delta} - c_s^2 \frac{k^2}{a^2} \delta = 0, \quad (4)$$

where the additional damping term $\Gamma(a)$ arises from the inter-sectoral relaxation dynamics derived in Companion X. The relation

$$\Gamma(a) = \alpha(a) H(a) \quad (5)$$

links the damping directly to the relaxation coefficient $\alpha(a)$, while the remaining terms follow from the Boltzmann hierarchy of Companion VII.

In the regime $\Gamma \sim \alpha H$, the growth of δ is suppressed by a factor that can be parametrised as an effective enhancement of the collapse threshold. Solving the equation in the relaxation-dominated regime yields

$$\delta(a) \propto a^{-\alpha/2}, \quad (6)$$

implying that the initial amplitude must be larger by a factor $a^{\alpha/2}$ to reach the standard collapse threshold at horizon re-entry.

Relaxation-Corrected Collapse Threshold

This motivates the parametrisation

$$\delta_c^{(A/B)} = \delta_{\text{std}} \left[1 + \beta \alpha_{A/B}(a) \right], \quad (7)$$

where $\beta = \mathcal{O}(1)$ captures the dependence on the perturbation profile. Because $\alpha_B(a) < \alpha_A(a)$ due to the lower temperature of sector B, the collapse threshold satisfies

$$\delta_c^{(B)} < \delta_c^{(A)}. \quad (8)$$

Exponential Sensitivity of PBH Formation

PBH formation is exponentially sensitive to the collapse threshold. Writing the PBH mass fraction explicitly,

$$\beta_{\text{PBH}}^{(A/B)}(M) \simeq \frac{1}{\sqrt{2\pi}} \frac{\sigma(M)}{\delta_c^{(A/B)}} \exp \left[-\frac{\delta_c^{(A/B)2}}{2\sigma^2(M)} \right], \quad (9)$$

makes clear that even a small relative change,

$$\frac{\Delta\delta_c}{\delta_c} \sim \beta \alpha_{A/B}(a), \quad (10)$$

induces an exponentially large difference in β_{PBH} .

This is the origin of the strong sectoral PBH asymmetry in RDCC:

$$\beta_{\text{PBH}}^{(B)} \gg \beta_{\text{PBH}}^{(A)}.$$

The visible sector is exponentially suppressed, while the CPT sector naturally forms PBHs.

5 Thermodynamics and Dynamics of CPT-Sector PBHs

PBHs formed in the CPT-conjugate sector inherit the thermodynamic and dynamical properties of that sector. Three features are particularly important:

5.1 Temperature Dependence

The colder temperature of sector B implies:

- a slower expansion rate,
- a larger horizon mass,
- reduced pressure support,
- a lower collapse threshold.

These effects combine to enhance PBH formation relative to the visible sector.

5.2 Relaxation History

The relaxation coefficient satisfies

$$\alpha_B(a) < \alpha_A(a), \quad (11)$$

implying reduced damping of super-horizon perturbations in sector B. This allows overdensities to grow more efficiently prior to horizon re-entry.

5.3 Mass Spectrum and Abundance

The PBH mass spectrum in sector B is shifted toward higher masses due to the larger horizon mass:

$$M_{\text{PBH}}^{(B)} \approx 4M_{\text{PBH}}^{(A)}. \quad (12)$$

The abundance is exponentially enhanced:

$$\beta_{\text{PBH}}^{(B)} \propto \exp \left[-\frac{\delta_c^{(B)2}}{2\sigma^2} \right], \quad \delta_c^{(B)} < \delta_c^{(A)}. \quad (13)$$

Together, these features establish PBHs as natural CPT-sector objects in RDCC, arising from the interplay between relaxation dynamics, sectoral thermodynamics, and the bipartite global state.

6 Observational Signatures

The sectoral PBH asymmetry predicted by RDCC leads to a number of distinctive observational signatures. These signatures arise from the suppression of PBHs in the visible sector, the enhanced PBH abundance in the CPT-conjugate sector, and the shift of the PBH mass spectrum toward higher masses in sector B.

6.1 Suppressed PBH Abundance in the Visible Sector

The relaxation-induced enhancement of the collapse threshold in sector A,

$$\delta_c^{(A)} = \delta_{\text{std}} [1 + \beta \alpha_A(a)], \quad (14)$$

leads to an exponential suppression of PBH formation. This prediction is consistent with microlensing, CMB, and SGWB constraints.

6.2 Enhanced PBH Abundance in the CPT Sector

Because $\alpha_B(a) < \alpha_A(a)$, the collapse threshold in sector B is lower,

$$\delta_c^{(B)} < \delta_c^{(A)}, \quad (15)$$

leading to an exponentially larger PBH abundance. These PBHs are not directly observable, but their existence is a robust prediction of the bipartite global state.

6.3 Sectoral Mass Asymmetry

The colder temperature of sector B implies a larger horizon mass at collapse,

$$M_{\text{PBH}}^{(B)} \approx 4M_{\text{PBH}}^{(A)}, \quad (16)$$

leading to heavier PBHs in the CPT sector.

6.4 Connection to Tensor Modulation (Companion II)

The sectoral PBH asymmetry is part of a broader pattern of inter-sectoral imprints in RDCC. Companion II shows that the same bipartite structure and relaxation dynamics induce a modulation of the tensor sector, affecting the primordial gravitational-wave background. Both effects — the CPT-sector PBH population and the tensor modulation — arise from the same underlying inter-sectoral coupling and the temperature asymmetry between the sectors. PBHs therefore provide a complementary probe of the same physics that shapes the tensor sector in RDCC.

6.5 Gravitational-Wave Signatures

PBHs contribute to the stochastic gravitational-wave background (SGWB) through mergers and collapse. The suppression in sector A implies:

- reduced merger rate,
- no PBH-induced SGWB in the visible sector,
- possible indirect signatures from sector B.

6.6 Microlensing Constraints

The strong suppression of PBHs in sector A implies that microlensing surveys should observe no significant PBH population — consistent with current data.

6.7 Modified PBH Mass Spectrum

The relaxation-corrected mass function,

$$\beta_{\text{PBH}}^{(A)}(M) = \exp \left[-\frac{\delta_{\text{std}}^2 (1 + \beta \alpha_A(a))^2}{2\sigma^2(M)} \right], \quad (17)$$

predicts a suppressed and shifted mass spectrum in sector A.

6.8 Consistency with RDCC Temperature Asymmetry

The sectoral PBH asymmetry provides an independent test of the temperature ratio $T_B/T_A \approx 0.5$ derived in Companion X.

7 Forecasts

The sectoral PBH asymmetry predicted by RDCC provides multiple observational avenues for testing the framework. Future surveys will be able to probe the PBH sector through gravitational waves, microlensing, and the matter power spectrum.

7.1 Gravitational-Wave Forecasts

Space-based interferometers such as LISA, DECIGO, and BBO will be sensitive to the stochastic gravitational-wave background generated by PBH mergers in the CPT sector. While these PBHs are not directly observable, their merger history may leave detectable imprints in the SGWB.

7.2 Microlensing Forecasts

Microlensing surveys will continue to tighten constraints on PBHs in the visible sector. The RDCC prediction of strong suppression in sector A is consistent with current data and will be further tested by future surveys such as Roman and LSST.

7.3 Large-Scale-Structure Forecasts

The absence of PBH-induced anomalies in the matter power spectrum provides an additional test of the RDCC PBH sector. Future surveys such as Euclid and CMB-S4 will be able to probe these signatures with high precision.

8 Conclusion

In this Companion we have developed the primordial black hole (PBH) sector of the Relaxation-Driven Cyclic Cosmology (RDCC), introducing a new interpretation in which PBHs arise naturally in the CPT-conjugate sector. The colder temperature, reduced relaxation-induced damping, and altered perturbation dynamics in sector B enhance PBH formation, while the visible sector experiences a strong suppression due to the elevated collapse threshold and reduced horizon mass.

These effects arise directly from the bipartite global state and the universal relaxation mechanism introduced in Companion X. The resulting sectoral PBH asymmetry provides a powerful observational handle on the RDCC framework and yields concrete predictions for gravitational waves, microlensing, and the PBH mass spectrum.

With the development of the PBH sector, RDCC now offers a coherent and internally consistent description of the thermal history, matter content, and early-Universe dynamics across both sectors. PBHs emerge as a natural and robust component of the CPT sector and a key probe of relaxation-driven cosmology.

References

- [1] B. J. Carr and S. W. Hawking, *Mon. Not. Roy. Astron. Soc.* **168**, 399 (1974).
- [2] B. Carr and F. Kühnel, *Rev. Mod. Phys.* **93**, 015004 (2021).
- [3] M. Lehmann, “Relaxation-Driven Cyclic Cosmology,” RDCC Flagship Paper (2026).
- [4] M. Lehmann, “Companion X: Microscopic Derivation of the Relaxation Rate,” (2026).
- [5] M. Lehmann, “Companion XIII: The Neutrino and Dark Matter Sectors in RDCC,” (2026).